



**Thomas
Thwaites**









The world works not because a few people know a lot, but because many people know a little.

Economic complexity is about understanding how that knowledge comes together.



Economic complexity

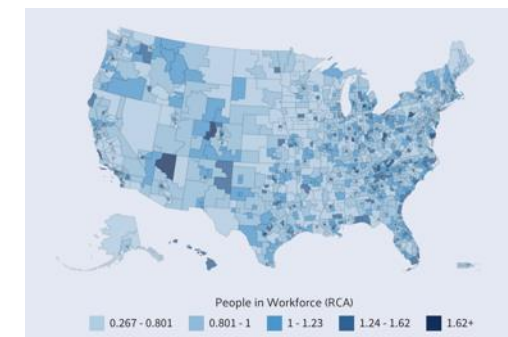
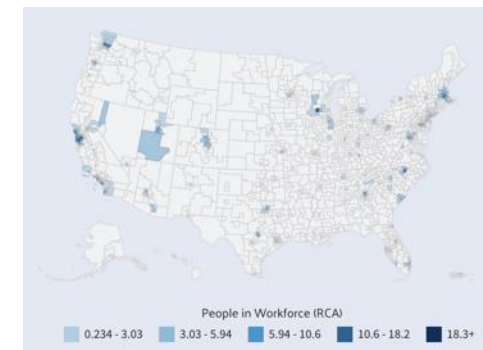
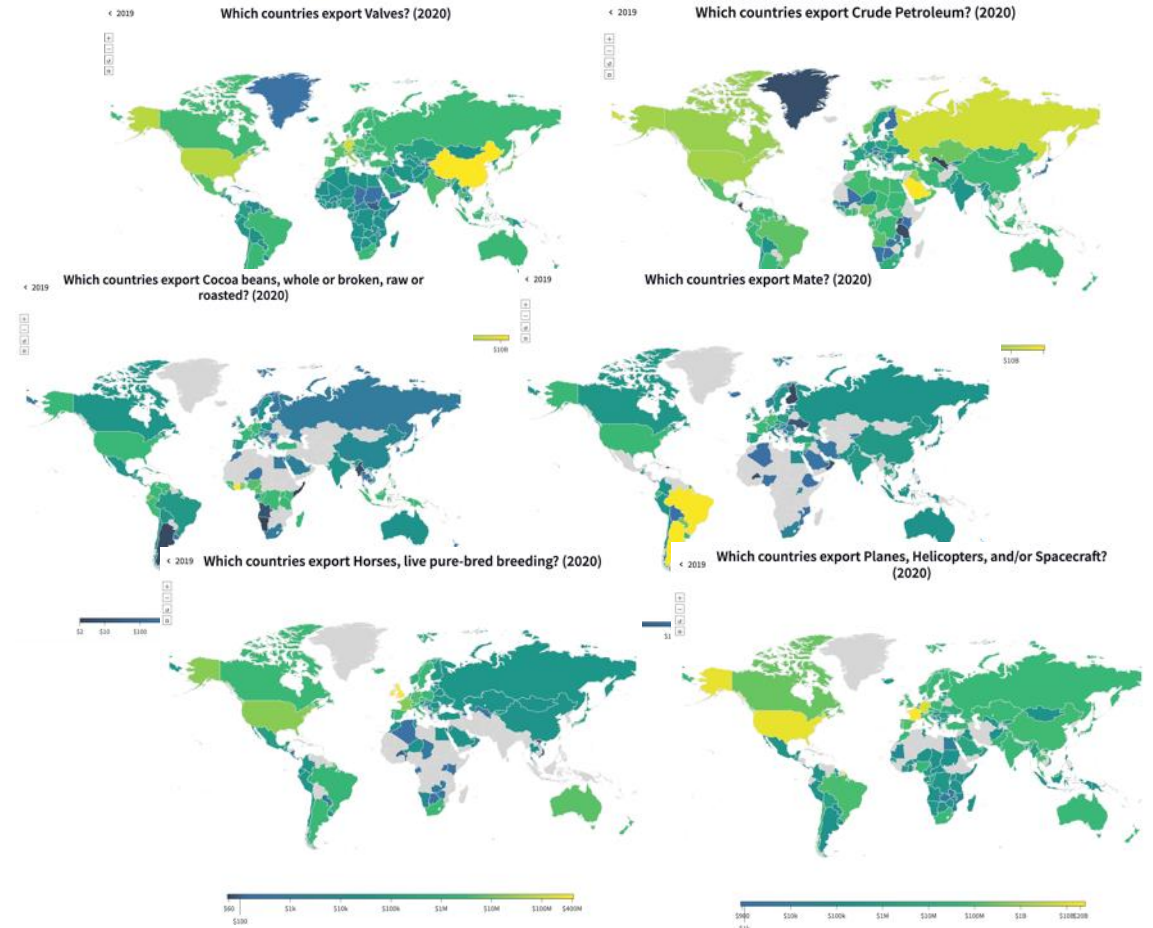
machine learning

+

economic data

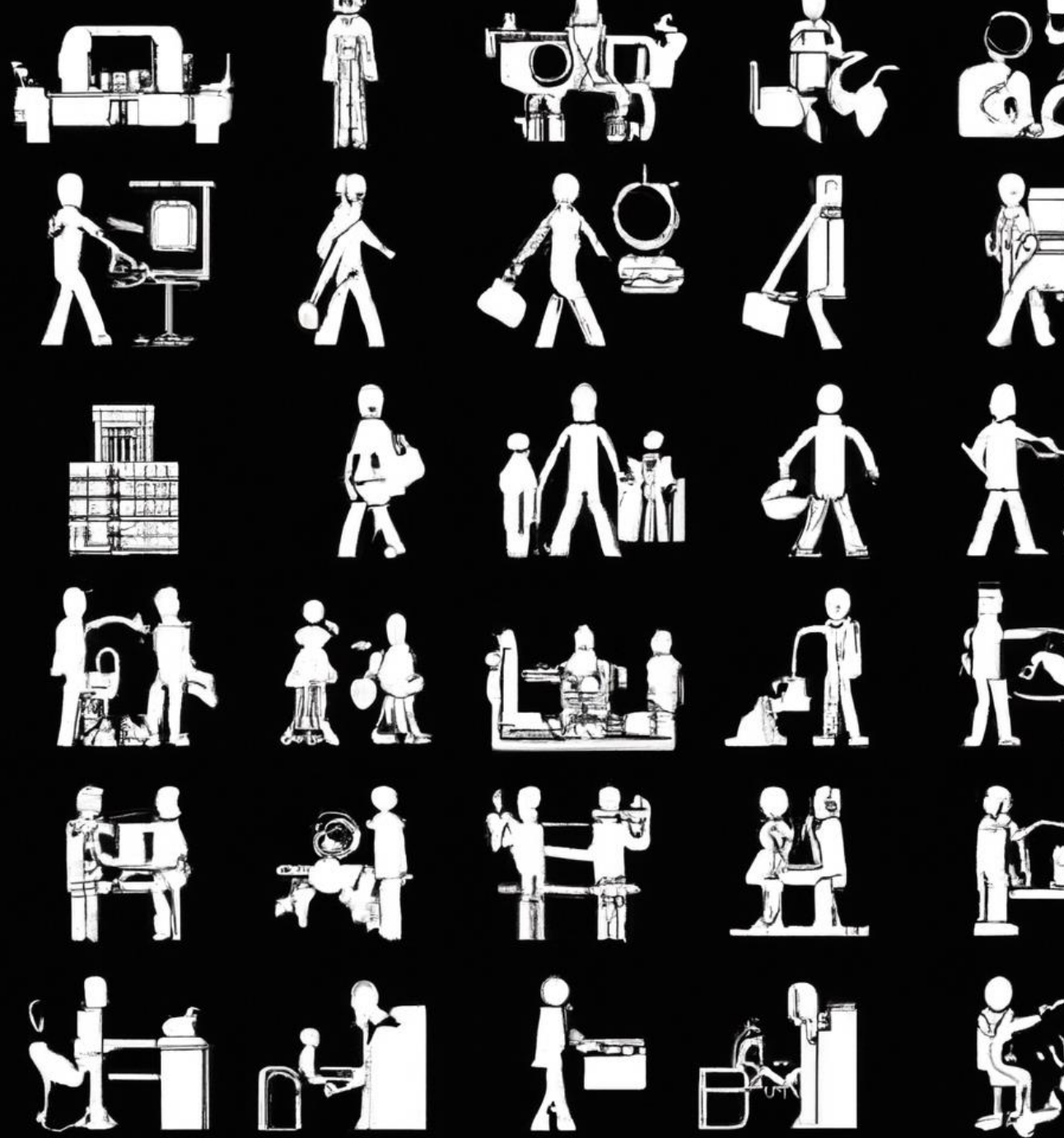
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development outcomes



Why Machine Learning?

Because factors of production,
and in particular **knowledge**,
are **non-interchangeable** or
non-fungible.



Non-Fungibility

Capital is fungible in theory. One dollar is the same as another dollar.

But physical capital and human capital (knowledge) are non-fungible in practice:

E.g. You cannot repurpose an oil refinery into a car manufacturing plant or replace a surgeon with a pianist.

Alphabet analogies

Letters and words are not interchangeable.

cat / cut
bat / bet
sit / set

Useful Analogies



Recipe analogies

Ingredients for a recipe or cocktail are not usually interchangeable



Non-fungibility requires changes of representation:

Parts-of-speech representation of words do not honor their non-fungibility

Verb, Nouns, Adjectives, and Adverbs List

Verbs	Nouns	Adjectives	Adverbs
accuse	accusation	accusing	accusingly
argue	argument	arguable	arguably
characterize	character	characteristic	characteristically
condition	condition	conditional	conditionally
darken	dark, darkness	dark, darkened	darkly
destroy	destruction	destructive	destructively
drink	drink, drunkenness	drunk, drunken	drunkenly

+ New chat

Phone models discussed.

AI Methods in Economics

Digital trade importance.

RQ: Papers, Ideas / T: Suggest

AI and Economics Revolution

Recommend TV Shows

New chat

New chat

Clear conversations

Upgrade to Plus NEW

Dark mode

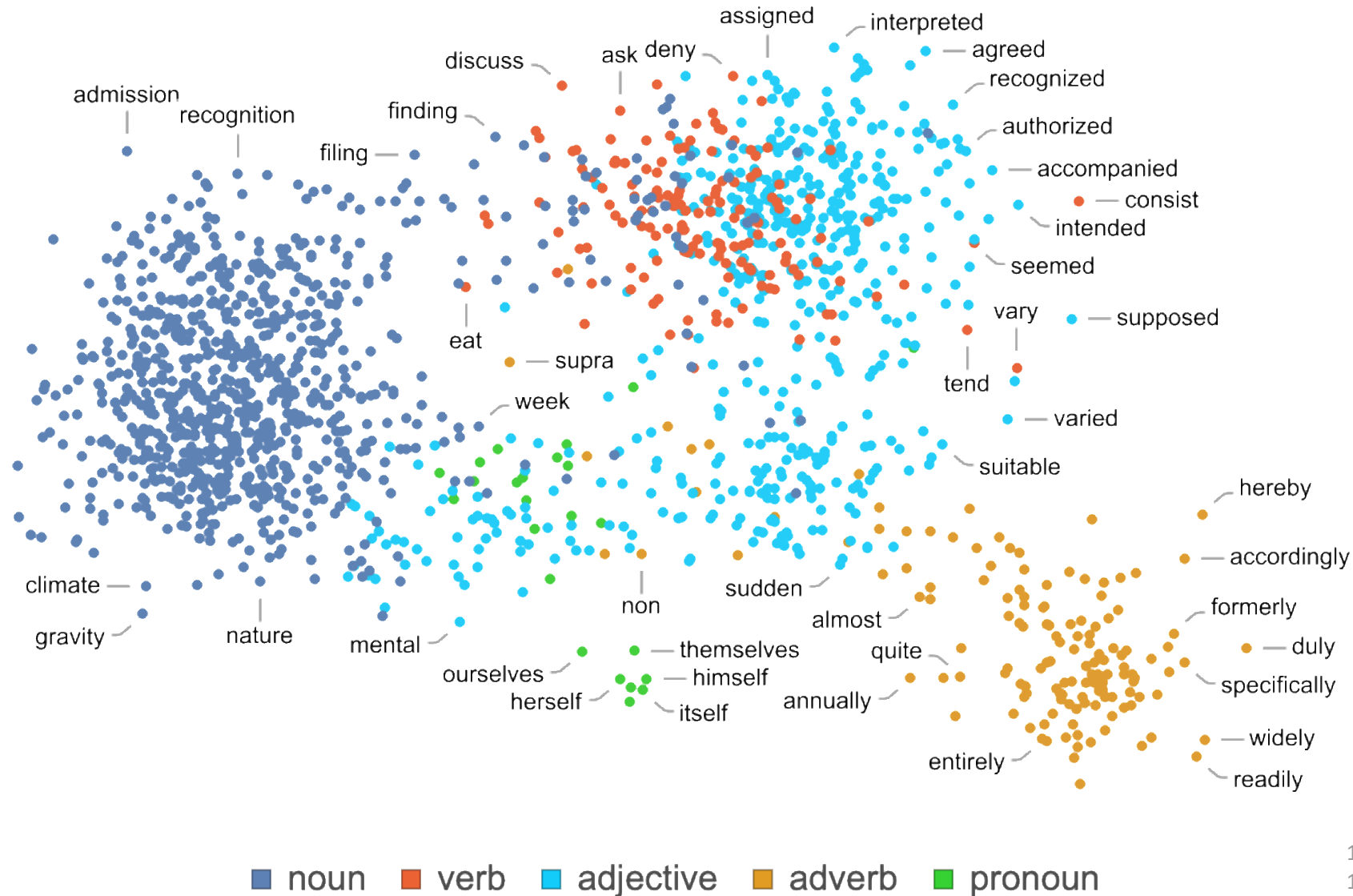
Updates & FAQ

Log out

ChatGPT

 Examples	 Capabilities	 Limitations
"Explain quantum computing in simple terms" →	Remembers what user said earlier in the conversation	May occasionally generate incorrect information
"Got any creative ideas for a 10 year old's birthday?" →	Allows user to provide follow-up corrections	May occasionally produce harmful instructions or biased content
"How do I make an HTTP request in Javascript?" →	Trained to decline inappropriate requests	Limited knowledge of world and events after 2021

Word Embeddings Provide Semantic Representations That Transcend Parts of Speech Limitations



“Parts of Speech” Representation of Products or Activities



Manufacture

Capital Intensive



Agriculture

Capital Intensive



Agriculture

Labor Intensive



Manufacture

Labor Intensive

AI or LLMs

Just like we can **count the number** of words in each sentence or paragraph, and their co-occurrences, to create representations of their semantic meaning, we can **count the number** of economic activities that are present across cities, regions, and countries to create representations of the knowledge embedded in them.

Economic Complexity



Spark Ignition Engines, Tobacco, Engine Parts, Aircraft Parts, Vaccines, Plywood, Tractors, Coffee, Frozen Bovine Meat, etc...



Spark Ignition Engines, Engine Parts, Aircraft Parts, Aircraft, Wheat, Wine, Perfumes, Vaccines, etc...



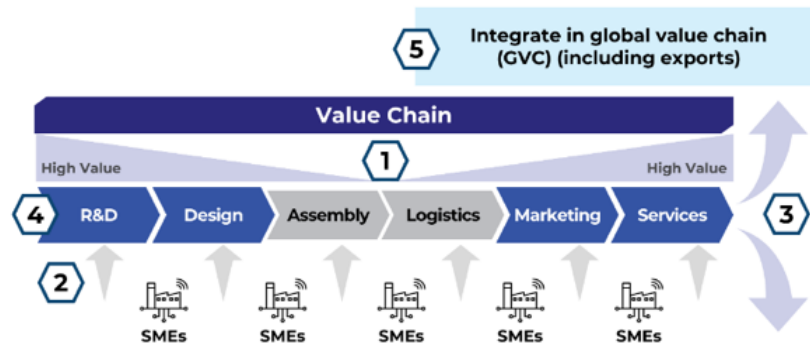
Crude Petroleum, Refined Petroleum, Petroleum Gases, Wheat, Aircraft Parts, etc.

But...
Who cares about
economic complexity?

Mission 1: Advance economic complexity

Mission 1 focuses on encouraging the industry to innovate and produce more sophisticated products to increase economic complexity

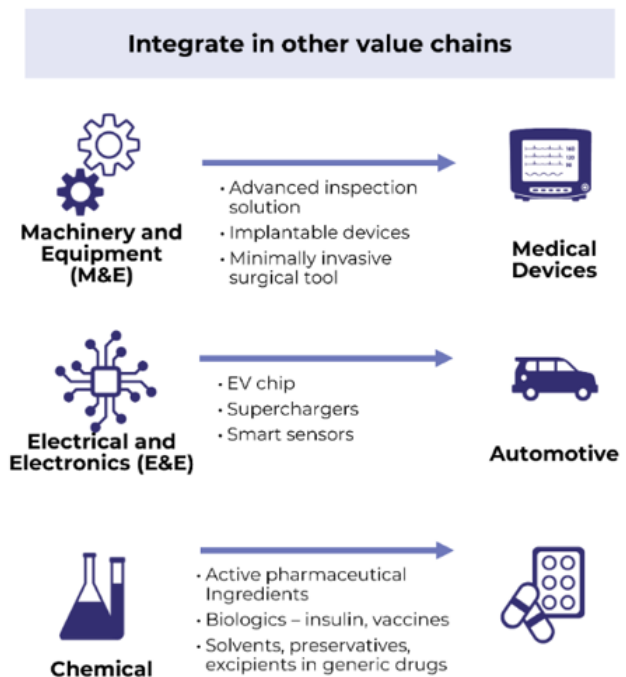
- 1 Expand to higher value-added activities
- 2 Develop ecosystem to support high value-added activities
- 3 Establish 'vertical integration' for GVC
- 4 Foster RDCI ecosystem
- 5 Increase manufacturing exports



Malaysia's strategic focus is on transitioning to higher value-added activities, moving beyond traditional manufacturing models towards an innovation-driven manufacturing hub.

This transformation involves fostering an ecosystem that encourages the growth of industries engaged in high-value economic activities, integrating value chains across sectors and promoting vertical integration among ASEAN countries.

The Research, Development, Commercialisation, and Innovation (RDCI) cycle plays a pivotal role in enhancing economic complexity and cultivating high-skilled talent, facilitating the introduction of innovative products and services that drive job creation and economic expansion. Central to this approach is the goal of increasing manufacturing exports to bolster Malaysia's economic growth and global competitiveness

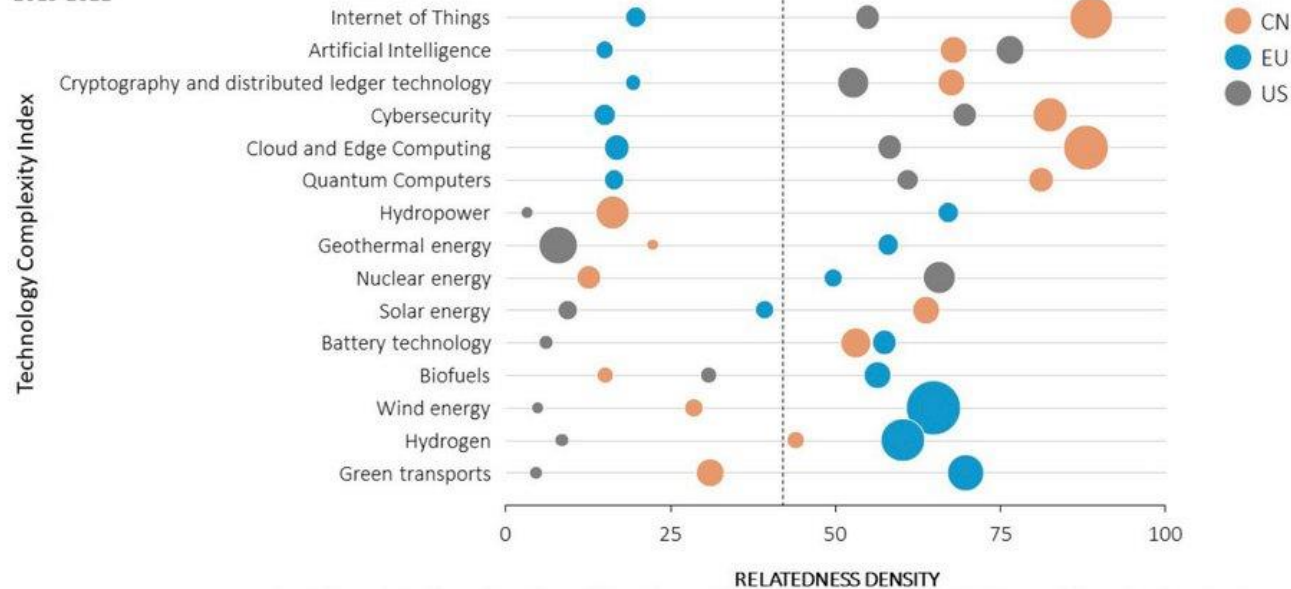


Economic Complexity is the First Mission of Malaysia's New Industrial Master Plan

FIGURE 2

The EU's position in complex (digital and green) technologies

2019-2022



Notes: The results are based on an analysis of patent data to understand the complexity and potential for specialisation in different technology areas. On the y-axis, technologies are ranked according to how advanced or complex they are, with scores ranging between 0 (less complex) and 100 (more complex). The x-axis (showing the relatedness density) represents how easily a country can build comparative advantage in a particular technology, depending on how closely related it is to other technologies the country is already strong in. The size of the bubbles shows how much each country has already specialised in a technology, using a measure of "revealed comparative advantage"(RCA), which reflects their competitive strenght in that field.

Source: European Commission, DG RTD.

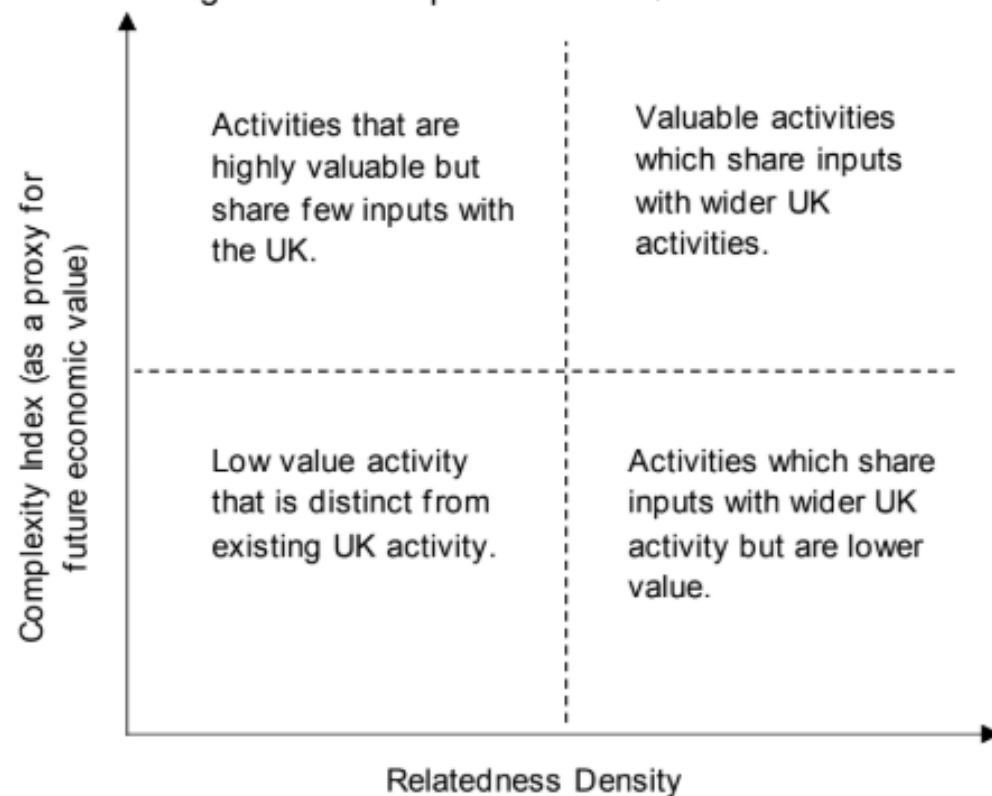


Economic Complexity was used in the recent Draghi Report to the EU

The future of European Competitiveness. "Draghi Report" 2024



Fig 1. – Smart Specialisation Quadrant Plot



In the United Kingdom



Government
Office for Science

Technology and Science Insights (TSI)

TSI(24)039

Version: 20th December 2024

Where is it possible to develop UK domestic productive capability in Emerging and Disruptive Technologies?

This is experimental work. There is good academic evidence that [economic complexity](#) is a significant predictor of long-term economic growth. This has been used to inform industrial policy by many national governments. This is the first time, to our knowledge, that this approach been used within HMG to compare a range of emerging technologies. This analysis has been conducted at pace to inform the Industrial Strategy and has not been subject to expert review.

Economic Complexity has motivated the creation of dozens of economic data observatories



OEC



COTEQ

LA NOBLE Y GRAN ECONOMÍA EN CHILE



DataMÉXICO



ES

Observatorio Económico



Observatorio
Institucional



HEAL & HY
Communities, NC



DATA USA:



ITP Producción



DATA AFRICA



DataChile

EXPLORA LOS ÚLTIMOS DATOS DE COMPLEJIDAD ECONÓMICA PARA ESPAÑA Y SUS PROVINCIAS

 Descubrir datos de España

Comenzar a explorar 

TOP PROVINCIAS

Ver fichas de España →



TOP PRODUCTOS

Ver fichas de Productos →

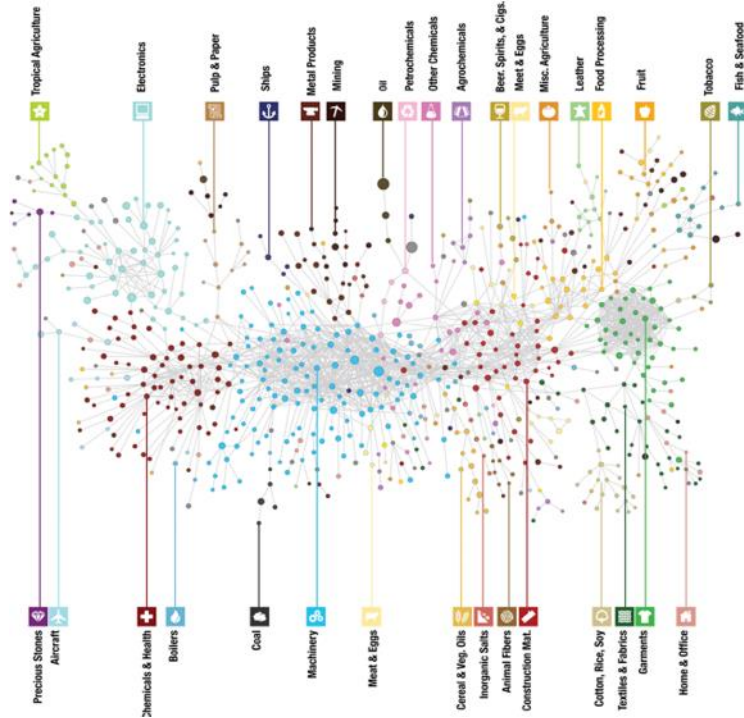


INDICADORES DE COMERCIO

Ver herramienta de comercio →

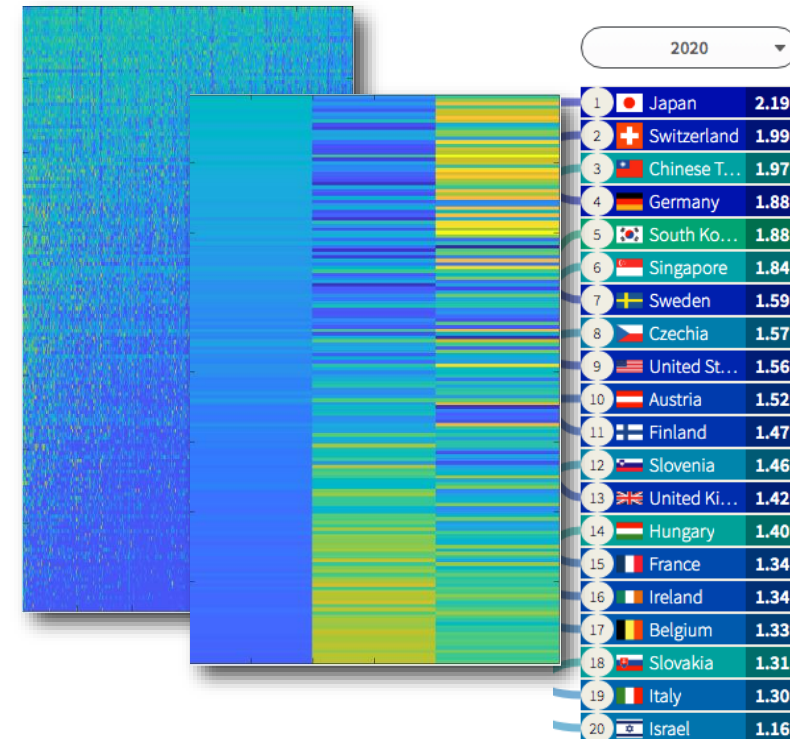
Two Key Concepts

Relatedness



Hidalgo et al. Science (2007)

Complexity Indexes



Hidalgo & Hausmann. PNAS (2009)

Proximities (in products)

Similarity among activities

 **Audio and Video
Recording
Accessories**

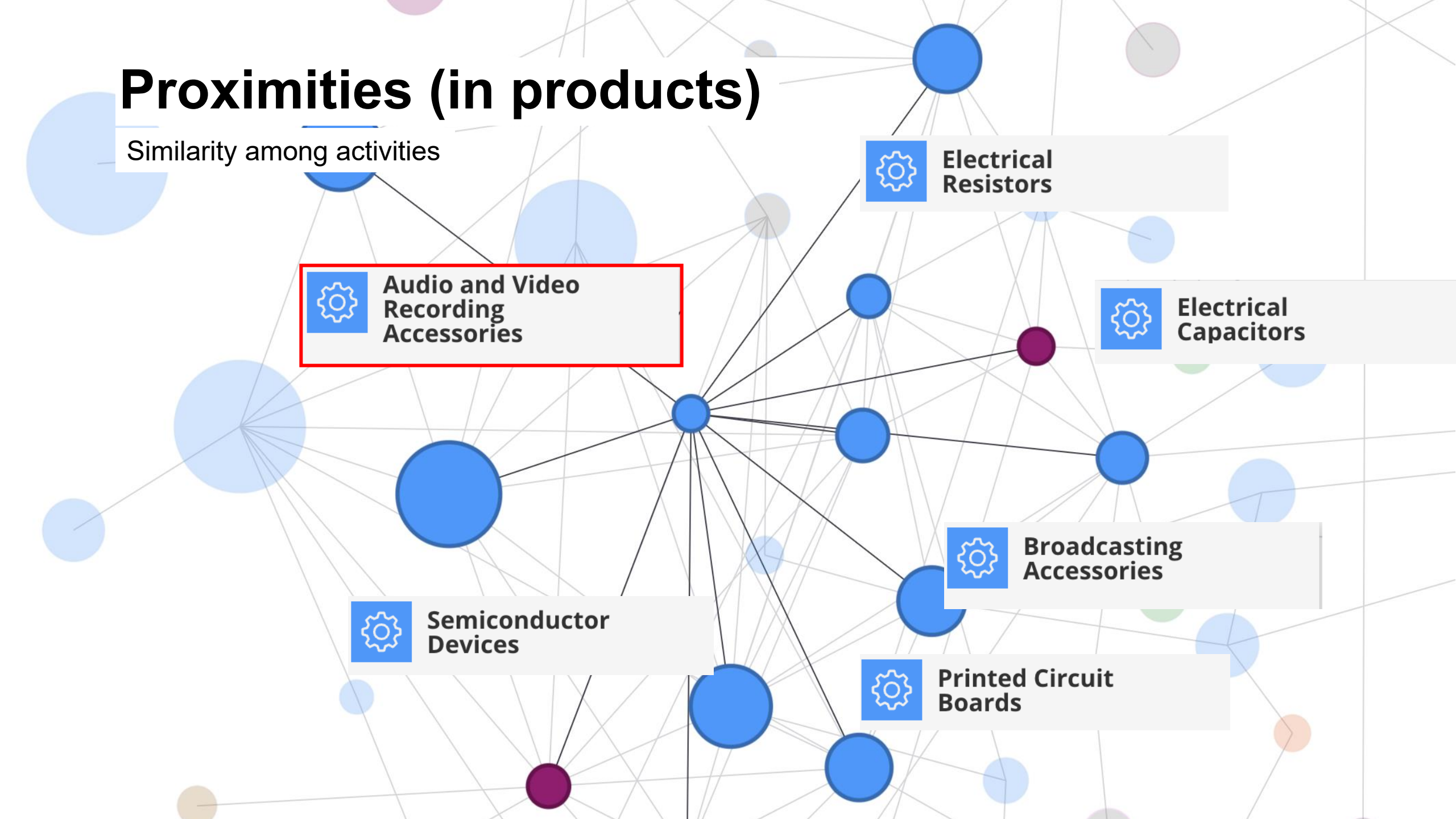
 **Electrical
Resistors**

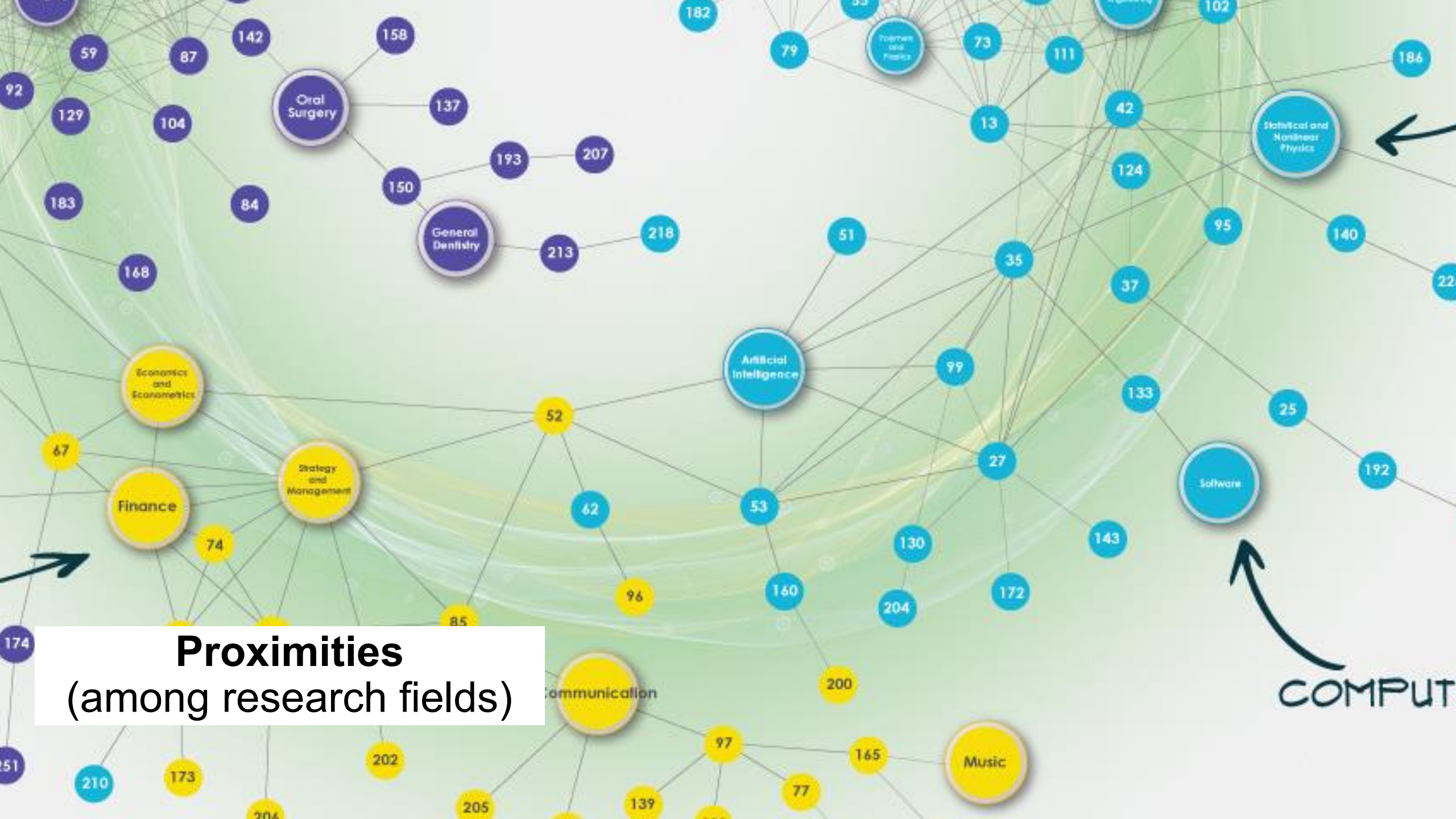
 **Electrical
Capacitors**

 **Broadcasting
Accessories**

 **Semiconductor
Devices**

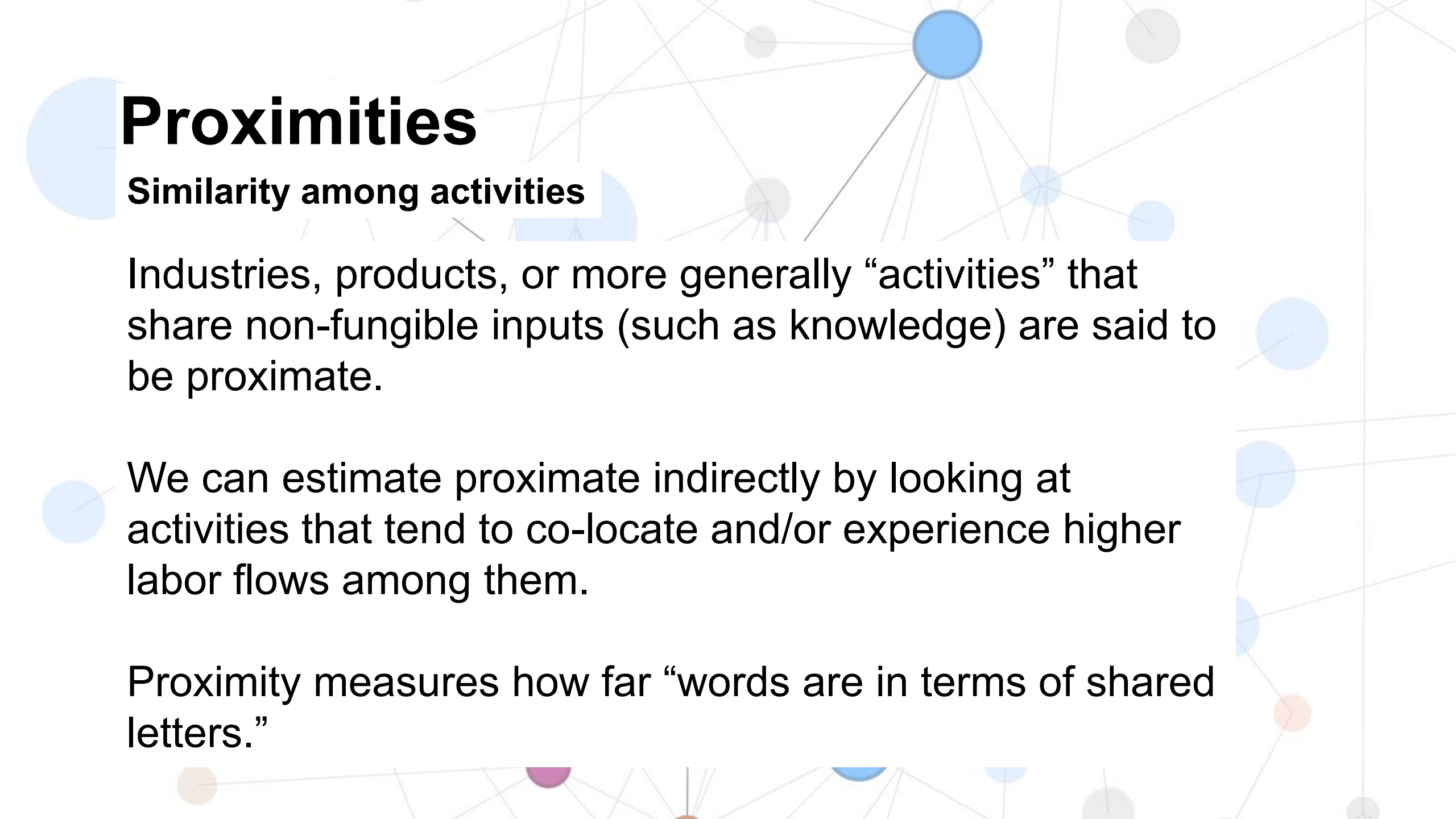
 **Printed Circuit
Boards**





Proximities
(among research fields)

COMPUT

A background network diagram with various colored nodes (blue, grey, orange, purple) connected by thin grey lines. The nodes are distributed across the slide, with a prominent blue node at the top center and others scattered throughout.

Proximities

Similarity among activities

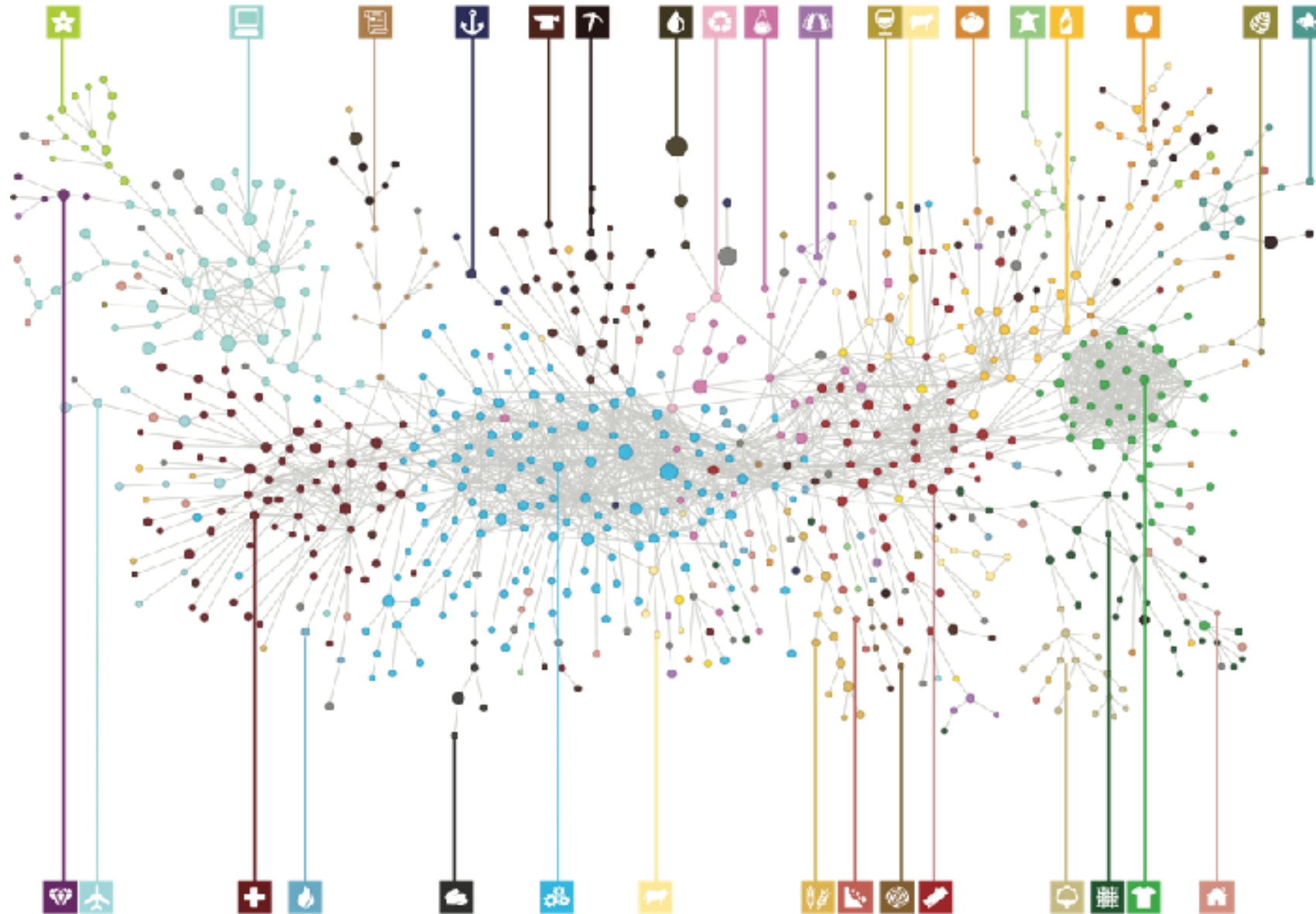
Industries, products, or more generally “activities” that share non-fungible inputs (such as knowledge) are said to be proximate.

We can estimate proximate indirectly by looking at activities that tend to co-locate and/or experience higher labor flows among them.

Proximity measures how far “words are in terms of shared letters.”

The Product Space

(co-exported products)



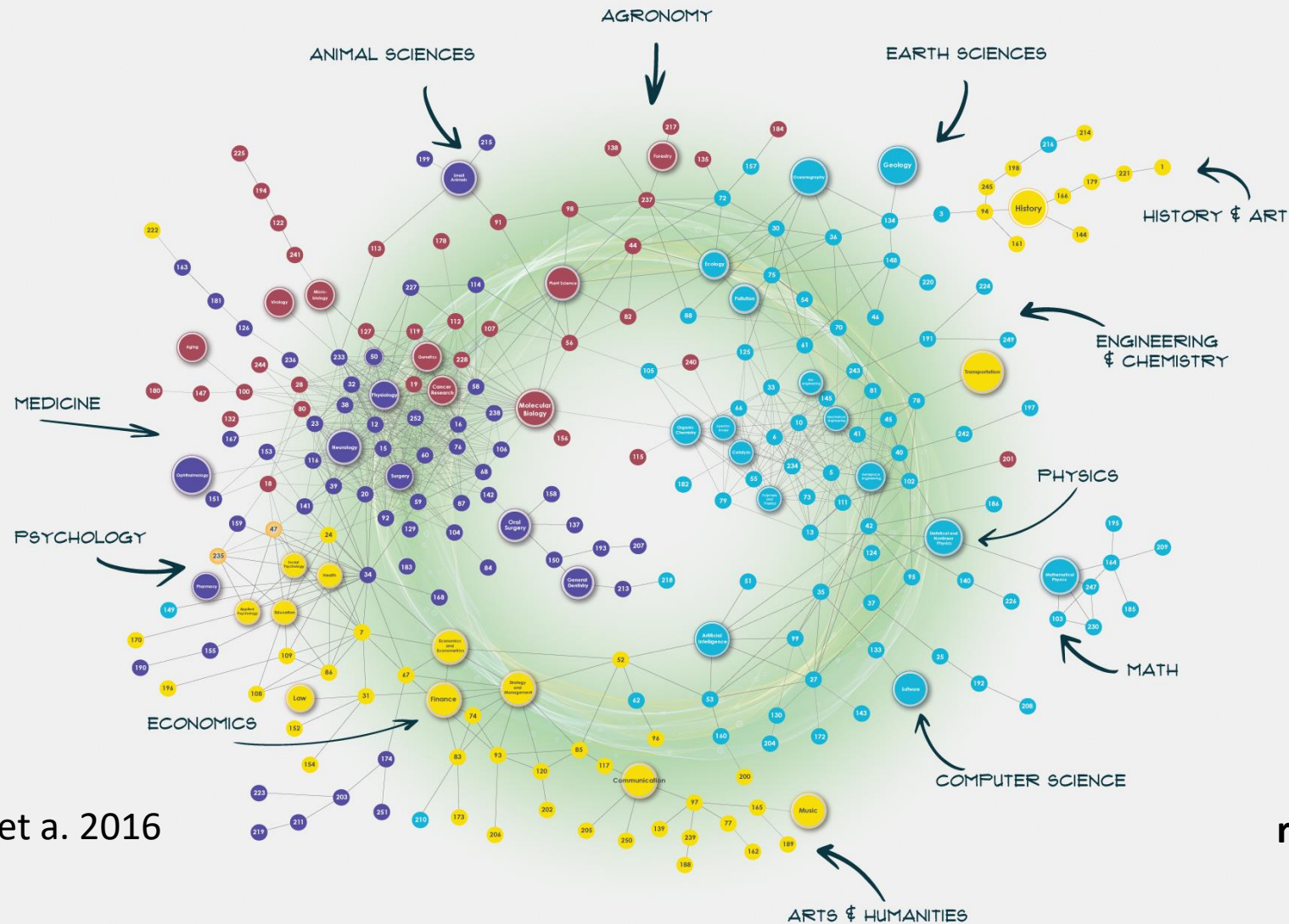
<https://oec.world>

Hidalgo et al. Science 2007

The Research Space 2025

Nodes represent subfields of research and are connected with other fields based on how likely an author is to publish in both of them

Life Sciences Health Sciences Social Sciences Physical Sciences

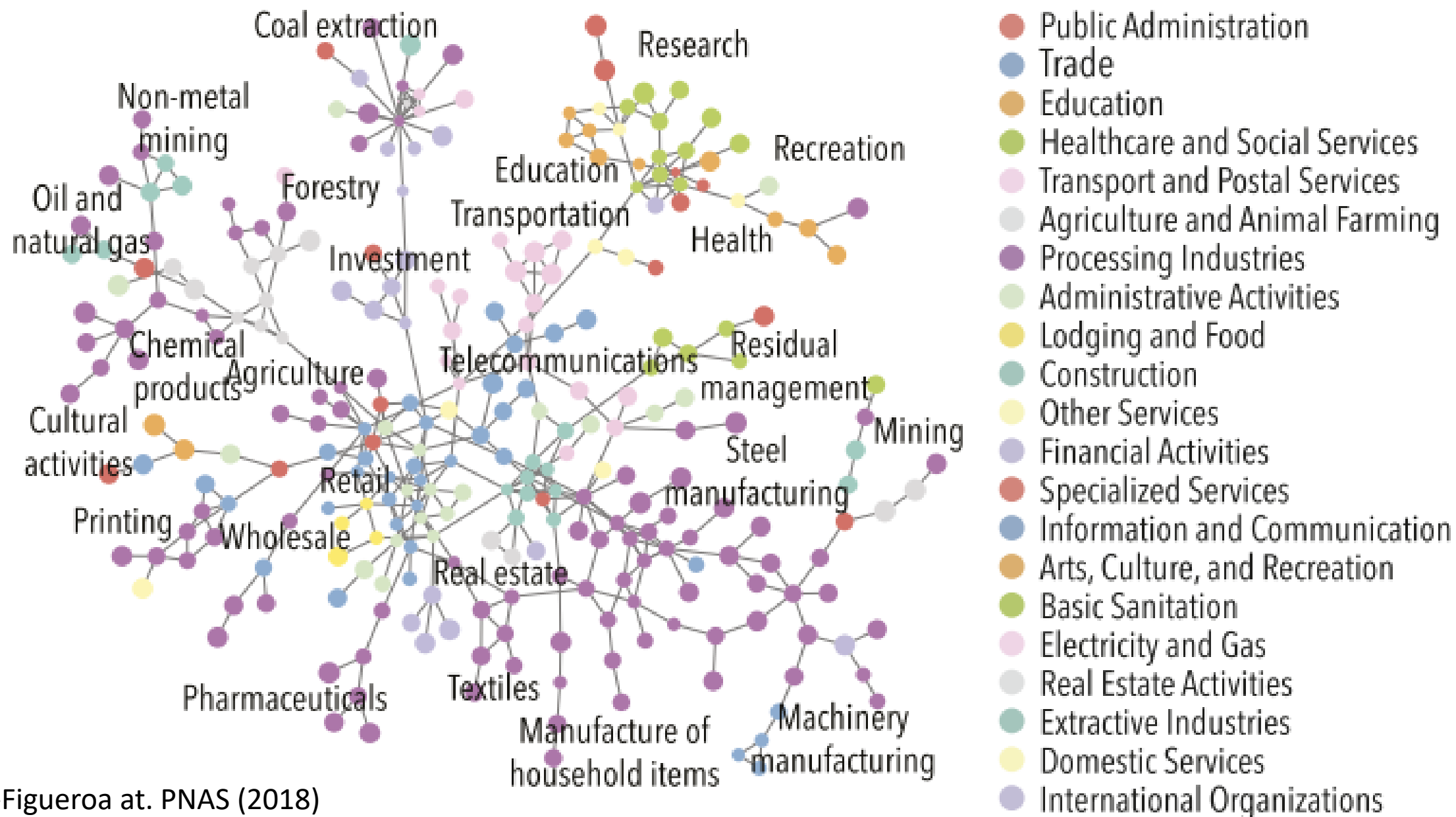


Guevara et al. 2016

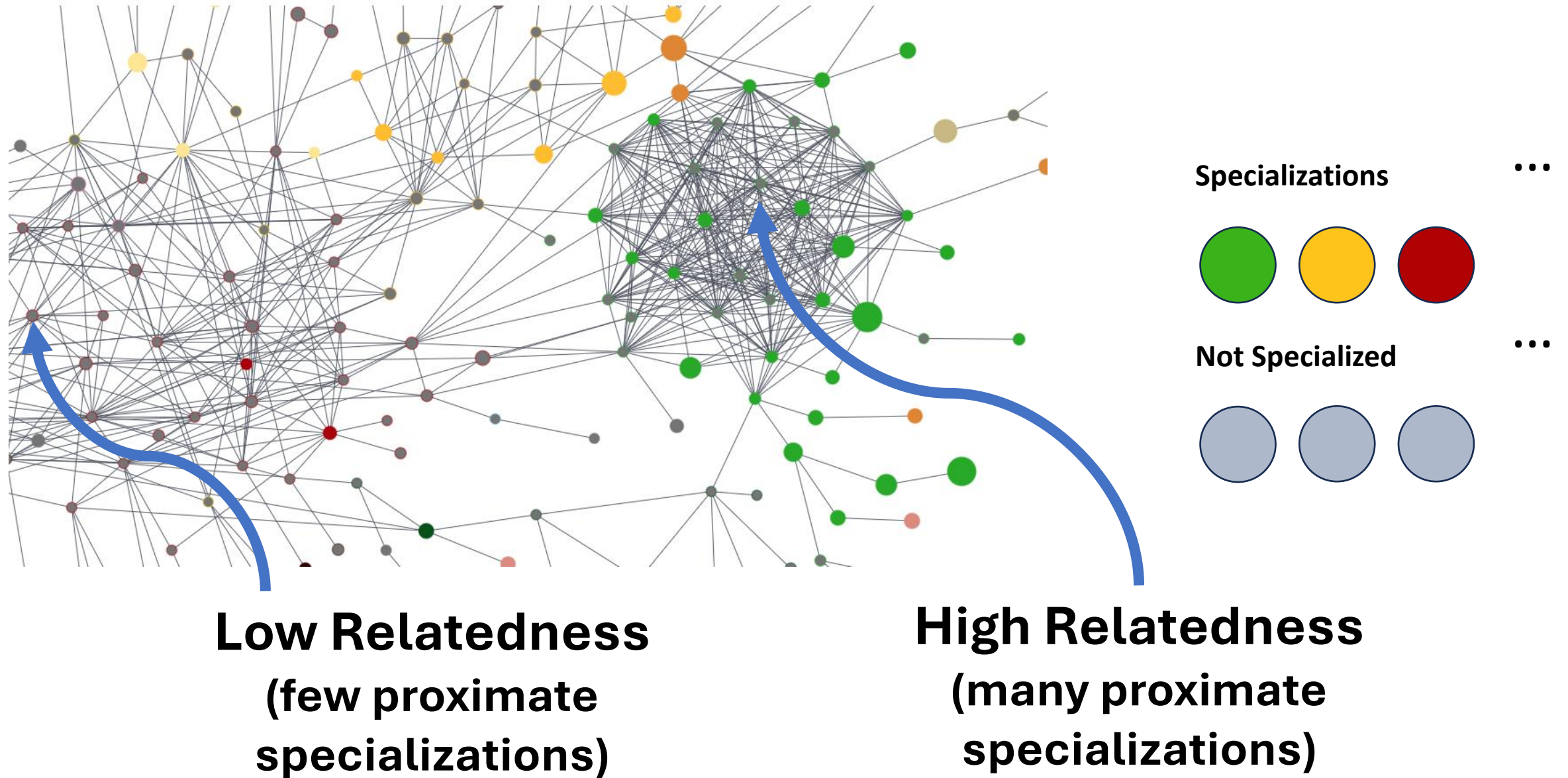
rankless.org

B

Industries



Relatedness



Relatedness

Relatedness provides an estimate of the latent capabilities available in an economy.

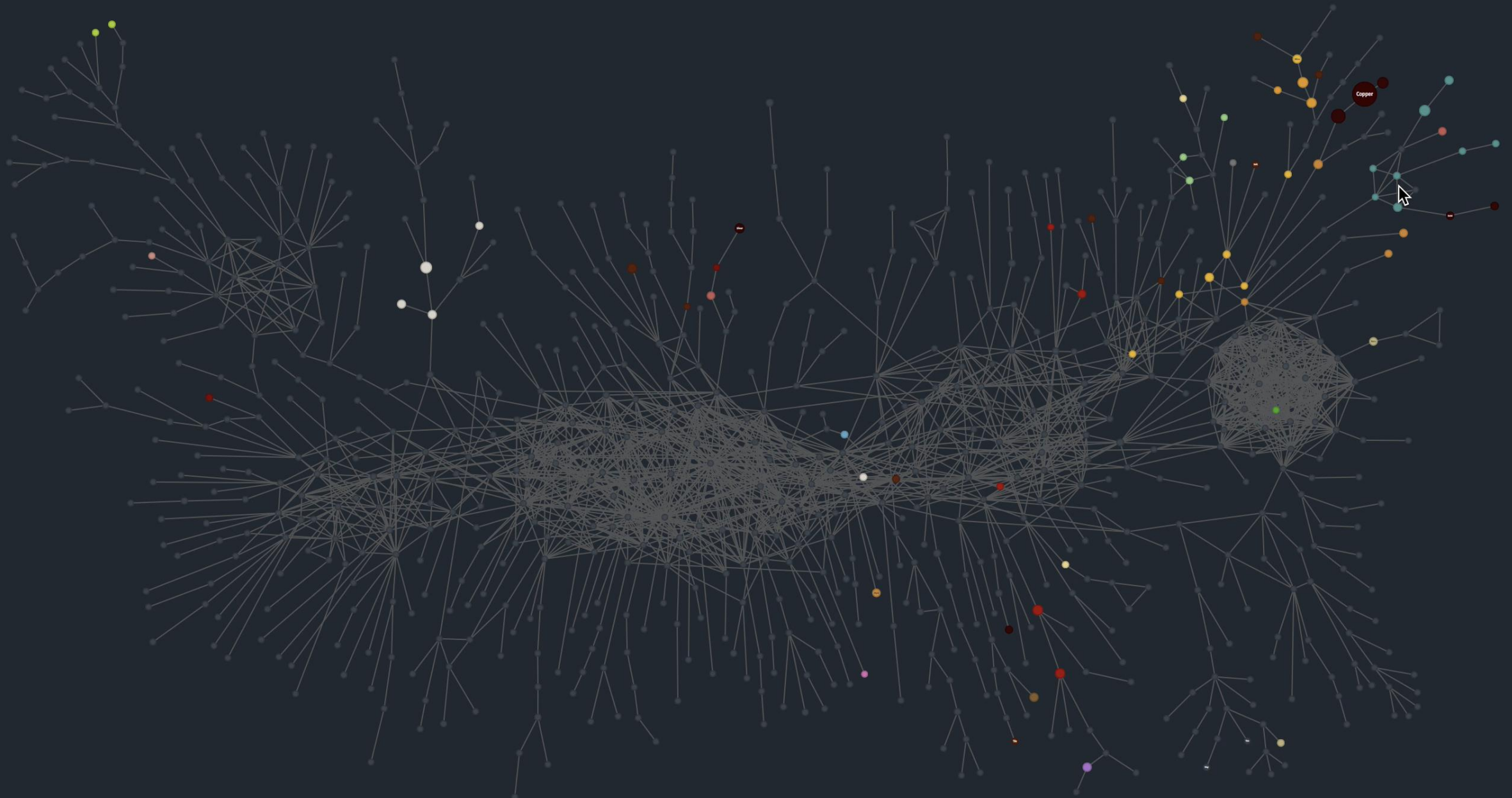
It measures the fit or affinity between an economy and an activity.

It is an estimate of the number of letters that an economy is missing to enter an activity.

DYNAMIC IMPLICATIONS:
Relatedness helps anticipate changes in economic structure!

What are the export opportunities of Chile? (1979)

TOTAL: \$3.67B

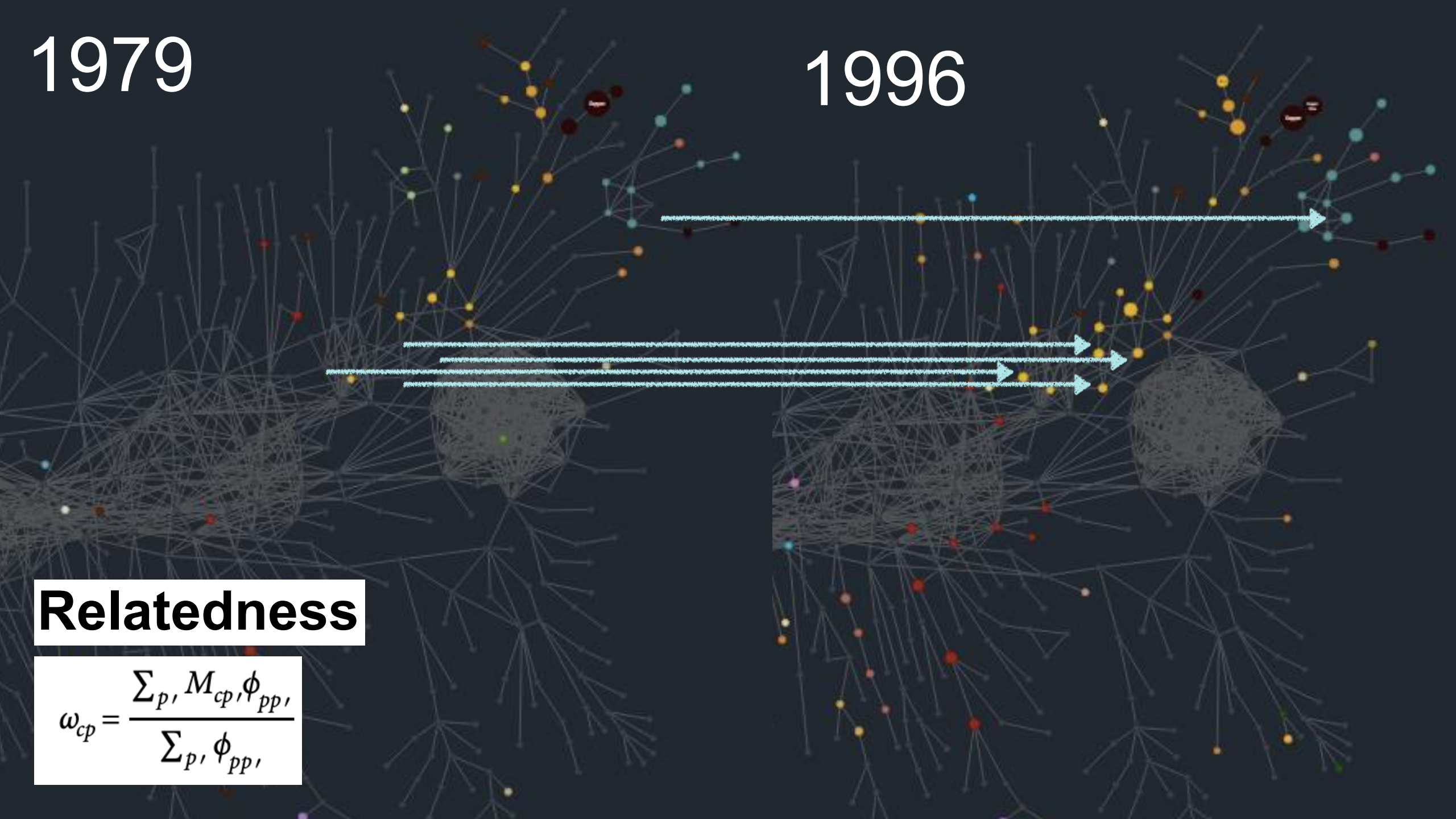


1979

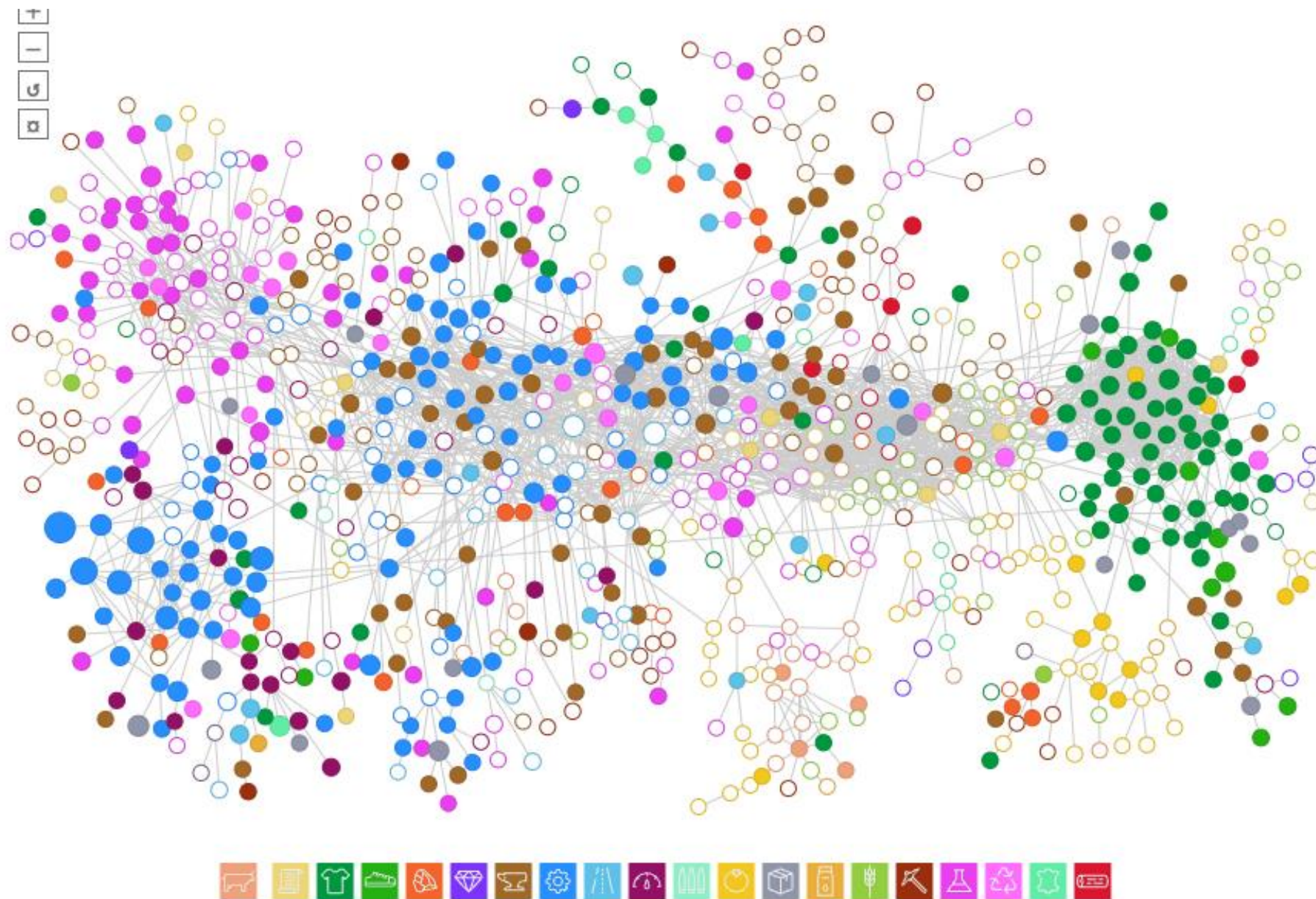
1996

Relatedness

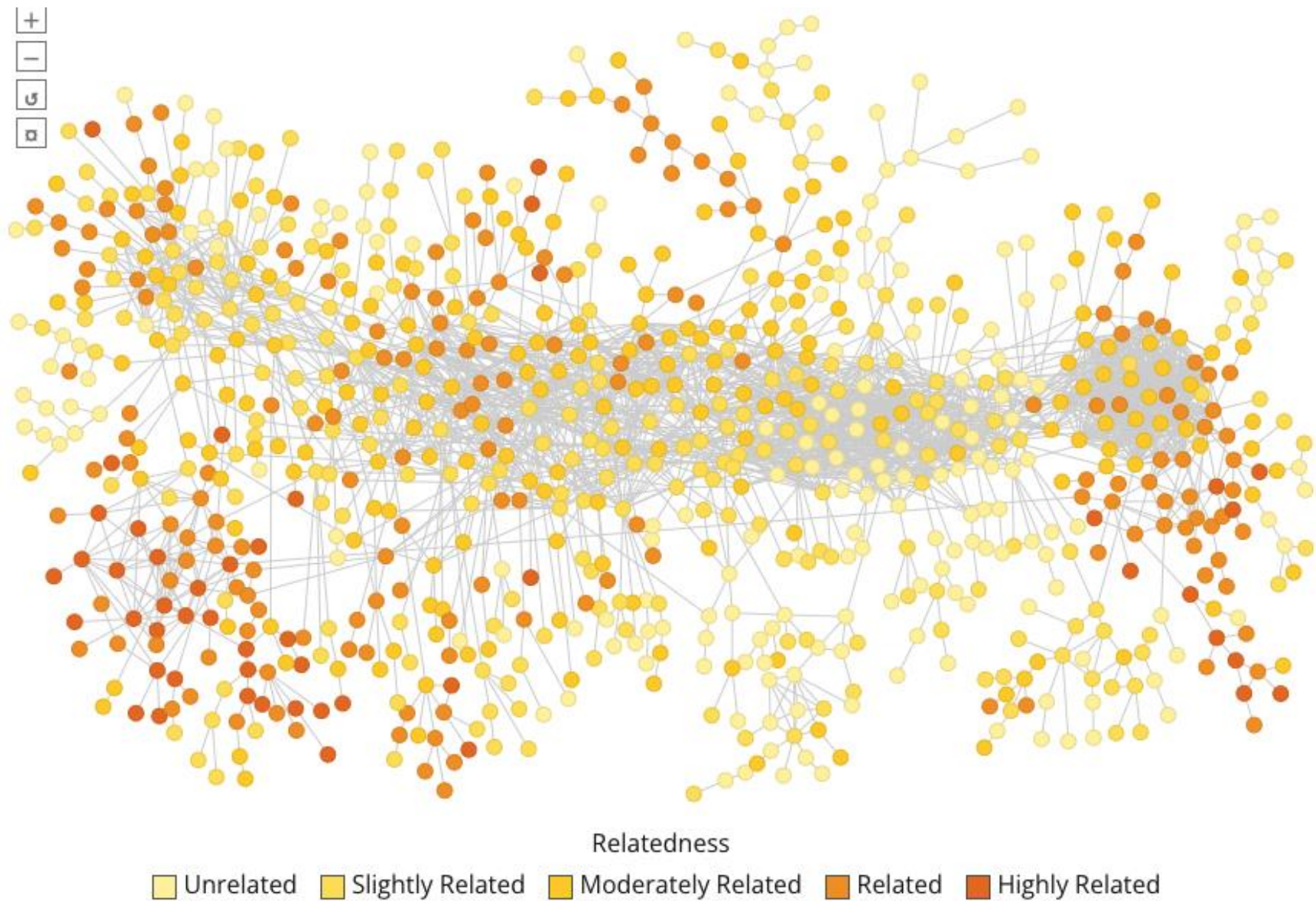
$$\omega_{cp} = \frac{\sum_{p'} M_{cp'} \phi_{pp'}}{\sum_{p'} \phi_{pp'}}$$



Product Space of China 2023



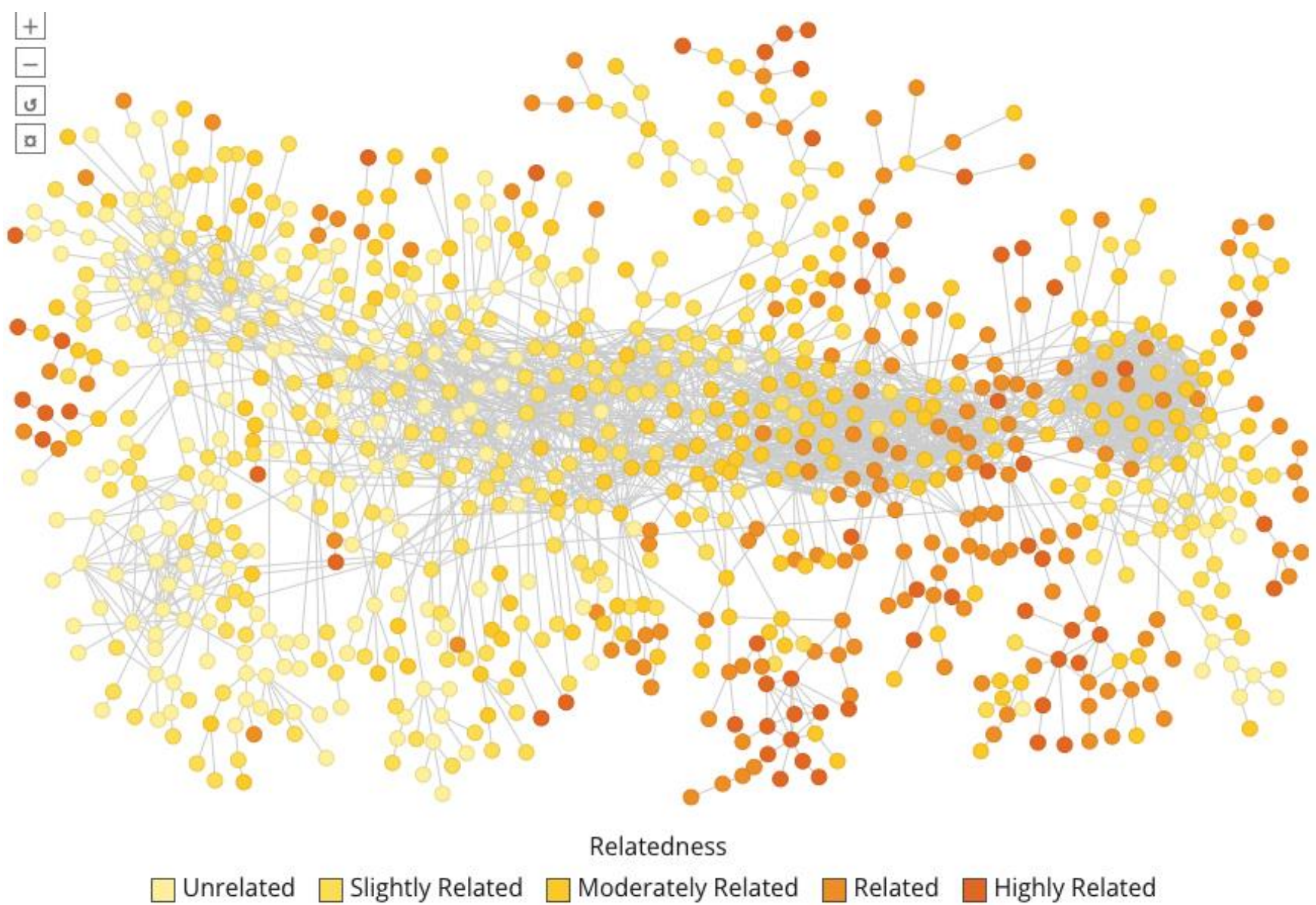
Relatedness of China 2023



Product Space of Chile 2023

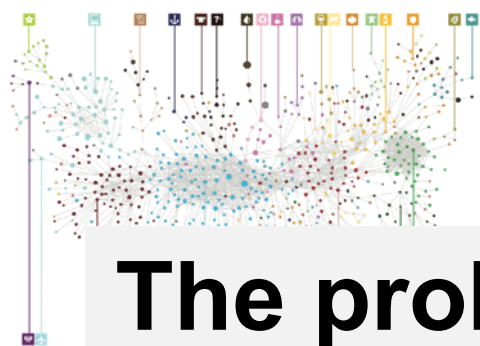


Relatedness of Chile 2023

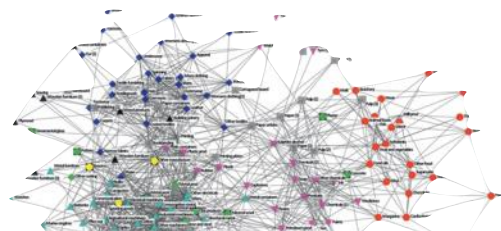


The Principle of Relatedness

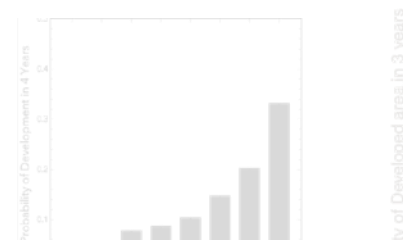
Products



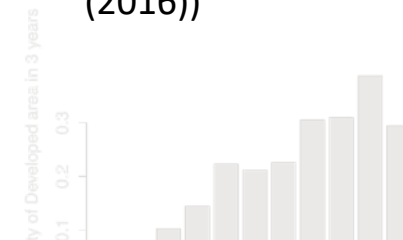
Industries



Products
(Hidalgo et al 2007)



Research Areas
(Guevara et al. (2016))



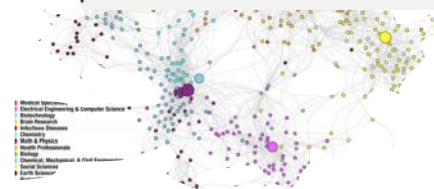
Patents

(Kogler et al. (2013), Boschma et al. (2015), Alstott et al. (2016))

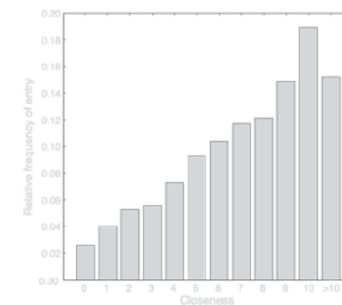
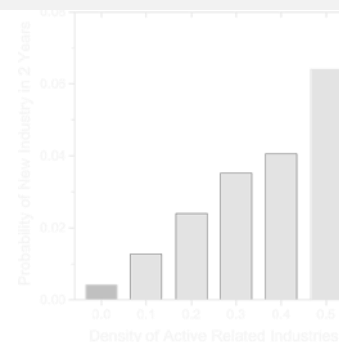
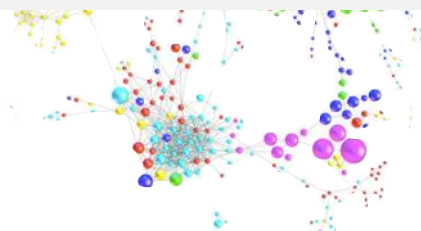


The probability that an economy enters (exits) an activity, increases with the number of related activities present in that location.

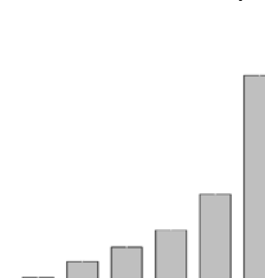
Research Areas



Patents



Gao et al. 2021



Game break!

?

PICK 5

🏆

👤

You are playing as a Guest User. Register in OEC to compete in the global leaderboard, coming soon. It is FREE! [SIGN IN NOW!](#)

🕒

You are playing a historical puzzle from 2025-12-07

Guess the 5 countries that export the most...



Cabbages

HS4 0704

🏆

0.00%

Pick an exporter

Submit

Play Pick-5 every day!

Built at Wed, 14 Jan 2026 13:36:16 GMT

Next in 09:55 hs

Game break!

?

PICK 5

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You are playing a historical puzzle from 2025-12-07

Guess the 5 countries that export the most...

Cabbages

HS4 0704

0.00%

Algeria

Germany

Niger

Nigeria

Ger

Submit

Play Pick-5 every day!

Built at Wed, 14 Jan 2026 13:36:16 GMT

Next in 09:55 hs

Game break!

?

PICK 5

🏆

👤

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🕒

You are playing a historical puzzle from 2025-12-07

Guess the 5 countries that export the most...

Cabbages

HS4 0704

🏆

2.59%

🇩🇪

1

🇩🇪

GERMANY

\$72.3M

9

Pick an exporter

Submit

Play Pick-5 every day!

Built at Wed, 14 Jan 2026 13:36:16 GMT

Next in 09:55 hs

Game break!

?

PICK 5

You are playing as a Guest User. Register in OEC to compete in the global leaderboard, coming soon. It is FREE! [SIGN IN NOW!](#)

You are playing a historical puzzle from 2025-12-07

Guess the 5 countries that export the most...

Cabbages

HS4 0704

12.07%

1

GERMANY

\$72.3M

9

2

NETHERLANDS

\$265M

5

Pick an exporter

Submit

Play Pick-5 every day!

Built at Wed, 14 Jan 2026 13:36:16 GMT

Next in 09:55 hs

Game break!

?

PICK 5

You are playing as a Guest User. Register in OEC to compete in the global leaderboard, coming soon. It is FREE! [SIGN IN NOW!](#)

You are playing a historical puzzle from 2024-05-22

Guess the 5 countries that export the most...

Perfumes

HS4 3303

0.00%

Pick an exporter

Pick

Link

<https://oec.world/en/games/pick-5?date=2024-05-22>



Game break!

?

PICK 5

🏆

👤

You are playing as a Guest User. Register in OEC to compete in the global leaderboard, coming soon. It is FREE! [SIGN IN NOW!](#)

🕒

You are playing a historical puzzle from 2026-02-08

Guess the 5 countries that export the most...

Cut Flowers

HS4 0603

🏆

0.00%

Pick an exporter

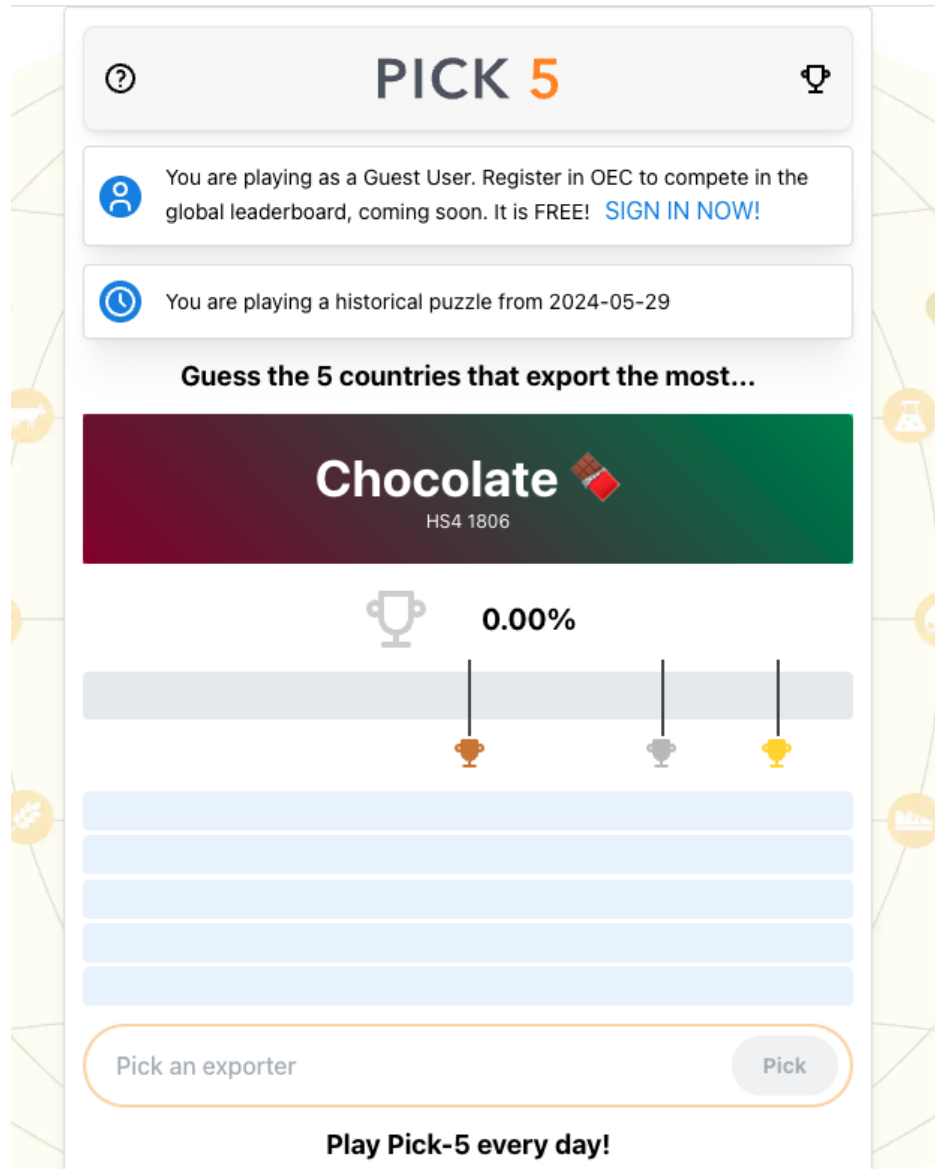
Pick

Link

<https://oec.world/en/games/pick-5?date=2026-02-08>



Game break!



Link

<https://oec.world/en/games/pick-5?date=2024-05-29>



Play Pick-5 every day!